

shipermansfriend

Version unspecified

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Status

Version 1.0.0-SNAPSHOT

Supported Platforms

- Android
- Windows
- Linux

About

Shiperman's Friend is a simple to use yet powerful ship spotter centric vessel tracker app. It has no fancy map view or vessel history - there are plenty of apps covering that kind of features. Instead, it focuses on the needs of a ship spotter in the field observing and photographing all the vessels out there.

Credits

iostream.io

First things first. Many thanks go out to the volunteer team at aisstream.io. Without your work to keep this up and running this project would not be possible or even exist!

National Anthems

Anthems were taken from [National Anthems Info](#)

National Flags

Flag icons were provided by flagsapi.com

Features

- Using [aisstream.io](#) as it's only data source as this the only service providing AIS livedata for free. It is mandatory to create a free account and obtain an api key (also free).
- Configurable double geo fence. Ship master data and safety messages are collected within the outer bounding box whereas position data is only collected within the inner bounding box. This inner bounding box is treated as the monitoring range of the ship spotter.
- Card view of all moving and moored vessels (separate tabs) in the monitored bounding box which shows the most important infos for a vessel.
- Vessel radar view to orientate in a dense environment which can be filtered by ship category or ship name, mmsi, imo number or callsign.
- Search tab to find and spot vessels of your interest.
- Bookmark vessels of interest for later export as CSV file.
- Mark vessels for alert when they enter the inner bounding box. This feature is using extrapolated positions to alert as early as possible. When a vessel alert is set it will also be bookmarked as alerts are only in memory and not being persisted.
- Safety message tab which display all safety relevant messages which have been sent in the outer bounding box.
- Keep track of message update frequency to determine if a moored vessel will depart soon.
- Play the anthem of the ships flag if available (237 worldwide countries supported). In part taken from [National Anthems Info](#).
- Export and import of collected ship masterdata so you can collect data overnight on your desktop and use this data on your mobile device.
- Export and import of settings. The API key is treated as a password and is stored encrypted with a platform key.
- Export
- To avoid any costs as most ship spotters are hobbyists no commercial data source is used to obtain extra infos like TEUs, etc. Unfortunately there is no free data source for this out there. I also do not want to scrape data out of websites as this would end in a legal mess. Instead, every vessel card has a button which sends you to the vessel page of [myshiptracking](#). As the search parameter the mmsi or imo code is used (if available). I think this is a good compromise and will lead to some extra traffic for that provider.

Downloads

This documentation is also available as [PDF Document](#)

You can obtain the zip file for desktop, the installer package for linux or the apk file for android from the artifacts section on the right side of this page.

Usage

Driving Vessels Tab

The tab shows all vessels currently under way within the warning range.

The top bar shows the coordinates of your current location. When you click on it you are being sent to googlemaps to show the location on a map.

Below you can see the amount of minutes ago of the last location update, the number of vessels in the view and the currently chosen inner radius.

The indicators on the right show the current network status with three indicators from left to right:

- Connectivity status: Green cost free connection available (lan or wlan), yellow costly connection available (cellular), red (disconnected)
- aisstream.io status: Green up, red down (using bottermilkgreens unofficial [monitor api](#))
- Current message status: Green (we receive ais messages), red we do not receive messages for a certain time.

The aisstream.io status up does not necessarily mean that we receive actually ais messages.

Under that the vessel cards are shown which contain only the information which can be obtained from aisstream.io. I decided not to scrape other webpages for additional interesting information like tonnage, TEUs etc. Instead, you can click on the middle button on the right to open a browser with the vessels page on [myshiptracking.com](#). I have chosen this provider as it is the only one which allows to search for mmsi numbers which is mandatory for this project as aisstream only sporadically provides imo numbers for vessels.

Rows

Vessel Name

Shows the vessel's name on the left (if available) and on the right two indicators:

- Heading with an arrow pointing to the direction
- Movement direction (in or outbound)

Country

The country flag and country name

Buttons

From left to right:

- Show vessels page on [myshiptracking.com](#)

- Show vessel location on google maps
- Show vessel location on radar page
- Play country anthem (if available)

Data Fields

All fields are determined either from a vessels master data or position data AIS messages. No data is scraped from other resources than aisstream.io.

- Distance in human-readable format
- Last message ago as formatted time period and in squared braces the average duration between messages (last three messages received). Higher frequency indicates mor active vessel. Especially useful info for moored vessels as the rise the message frequency drastically shortly before departure.
- Destination (if available). I try to decode the unlo code(s) for location of depart and location of destination. When the location of the destination is known from the internal port database this turns into a link which shows the position on google maps.
- Speed in knots and km/h
- Rate of turn in degrees per minute. Negative means vessel is turning left (indicated red on the radar), positive means vessel is turning right (indicated green on the radar).
- MMSI (Maritime Mobile Service Identity)
- Callsign (if available)
- IMO Number (International Maritime Organization Number) (if available)

When you click on the rightmost icon with the radar icon the radar page is opened with the selected vessel being highlighted with a white flashing circle.

The right box shows an icon for the ships category and size and draught (if available).

Shiperman's Friend			
53°32'45.52"N 9°58'4.96"E 00:06:54 57 20.0 km			
QUEEN Germany			
DISTANCE 1.8 km SPEED 7.5 Knots / 13.9 Km/h MMSI 218025480	LAST MESSAGE 0 00:02:30 [2] TURN RATE 0 °/min CALLSIGN DA3374	DESTINATION IMO	Passenger 63 x 12 m Draught 0.0 m
FLEETENKIEKER Germany			
DISTANCE 1.8 km SPEED 2.7 Knots / 5.0 Km/h MMSI 211375630	LAST MESSAGE 0 00:00:36 [2] TURN RATE CALLSIGN DJ7077	DESTINATION IMO	Passenger 26 x 6 m Draught 0.0 m
BIRGIT EHLERS Germany			
DISTANCE 2.0 km SPEED 5.3 Knots / 9.8 Km/h MMSI 211170960	LAST MESSAGE 0 00:06:34 TURN RATE CALLSIGN DG4596	DESTINATION Hamburg (DE) IMO	Passenger 17 x 5 m Draught 1.1 m
NEW YORK Germany			
DISTANCE 1.6 km SPEED 6.1 Knots / 11.3 Km/h MMSI 211149400	LAST MESSAGE 0 00:03:09 [0] TURN RATE CALLSIGN DH5926	DESTINATION Hamburg (DE) IMO	Passenger 19 x 3 m Draught 0.0 m

Driving Vessels Inbound Tab

The tab shows all vessels currently under way within the warning range and moving towards your current location (inbound traffic).

Shiperman's Friend

53°32'45.52"N 9°58'4.96"E
00:07:24 26 20.0 km

LOTSE 1 Germany → ⚙️ 📶 ▶️				Pilot 24 x 7 m Draught 0.0 m
DISTANCE 6.5 km SPEED 10.9 Knots / 20.2 Km/h MMSI 211281010	LAST MESSAGE 0 00:00:14 [1] TURN RATE 720 °/min CALLSIGN DA6505	DESTINATION Hamburg (DE) IMO		
ALTENWERDER Germany → ⚙️ 📶 ▶️				Passenger 30 x 8 m Draught 1.8 m
DISTANCE 7.3 km SPEED 9.2 Knots / 17.0 Km/h MMSI 211263850	LAST MESSAGE 0 00:03:14 [2] TURN RATE 720 °/min CALLSIGN DA8805	DESTINATION Hamburg (DE) IMO		
DLRG FOEHR WIED Germany → ⚙️ 📶 ▶️				Unknown
DISTANCE 19.4 km SPEED 3.4 Knots / 6.3 Km/h MMSI 211635080	LAST MESSAGE 0 00:01:15 TURN RATE 0 °/min CALLSIGN	DESTINATION IMO		
ALINA Germany → ⚙️ 📶 ▶️				Unknown 0 x 0 m Draught 0.0 m
DISTANCE 20.4 km SPEED 8.9 Knots / 16.5 Km/h MMSI 211324310	LAST MESSAGE 0 00:07:03 TURN RATE 0 °/min CALLSIGN	DESTINATION IMO		

Driving Vessels Outbound Tab

The tab shows all vessels currently under way within the warning range and moving away from your current location (outbound traffic).

Shiperman's Friend			
53°32'45.52"N 9°58'4.96"E 00:07:14 30 2 20.0 km			
MARLOU Netherlands DISTANCE 8.0 km SPEED 0.7 Knots / 1.3 Km/h MMSI 244660587 LAST MESSAGE 0 00:00:51 [1] TURN RATE CALLSIGN PD4501 DESTINATION Hamburg Hamburg Ter IMO Tanker 82 x 10 m Draught 1.8 m			
IDA RAMBOW Germany DISTANCE 13.7 km SPEED 11.1 Knots / 20.6 Km/h MMSI 218098000 LAST MESSAGE 0 00:02:37 [2] TURN RATE 0 °/min CALLSIGN DFHR2 DESTINATION Gdansk (PL) IMO 9354478 Cargo 149 x 23 m Draught 7.9 m			
SERRAHN QUEEN Germany DISTANCE 17.1 km SPEED 4.9 Knots / 9.1 Km/h MMSI 211643260 LAST MESSAGE 0 00:00:24 [3] TURN RATE CALLSIGN DC9466 DESTINATION Hamburg (DE) IMO Wing In Ground 34 x 6 m Draught 0.0 m			
KOELN I Germany DISTANCE 17.3 km SPEED 6.5 Knots / 12.0 Km/h MMSI 211191500 LAST MESSAGE 0 00:00:51 [1] TURN RATE 0 °/min CALLSIGN DD7509 DESTINATION Sandauhafen Hamburg/ IMO Cargo 173 x 11 m Draught 1.6 m			

Moored Vessels Tab

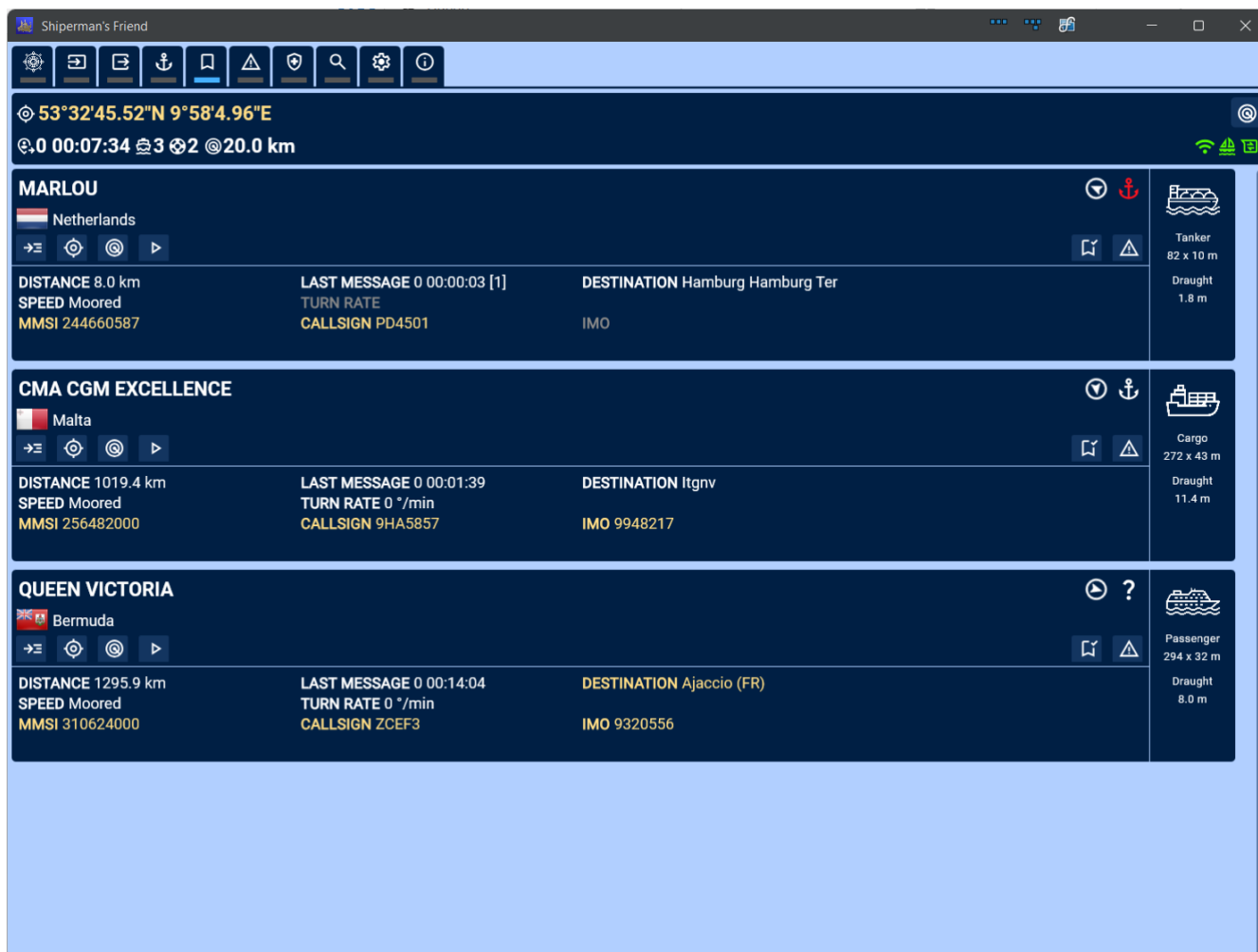
The tab shows all vessels with a sog (speed over ground) less than 0.5 knots. Those are considered as moored. The overall layout is the same as for the driving vessels tab.

Shiperman's Friend

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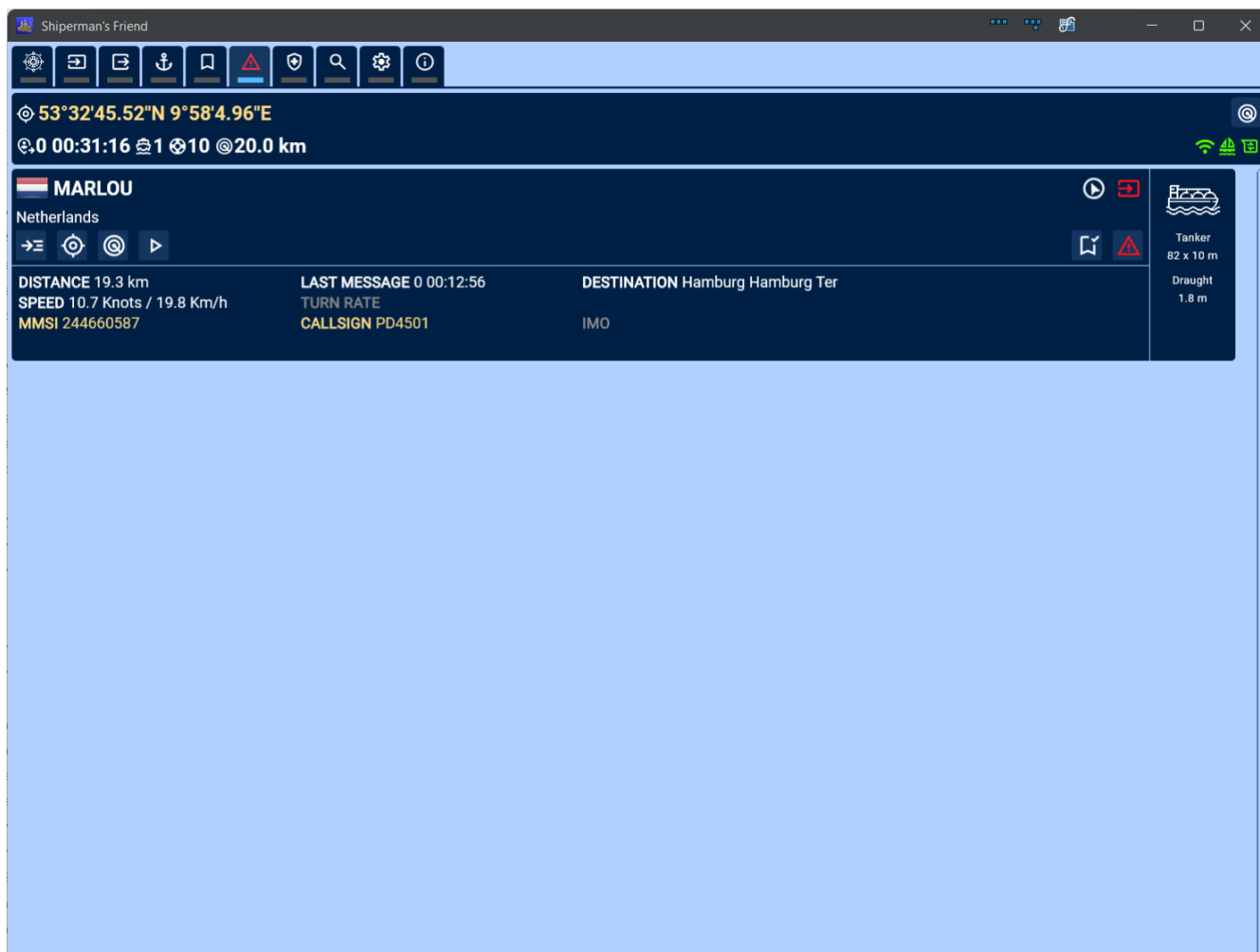
Starred Vessels Tab

This tab shows all vessels which you have bookmarked with the starred vessel button regardless if it is in range or not. When you bookmark a vessel this is indicated by a little checkmark on the bookmark icon. The current vessel data is then written to the database. The list of bookmarked vessels can be exported from the settings page. When the vessels have been exported the database table is cleared.



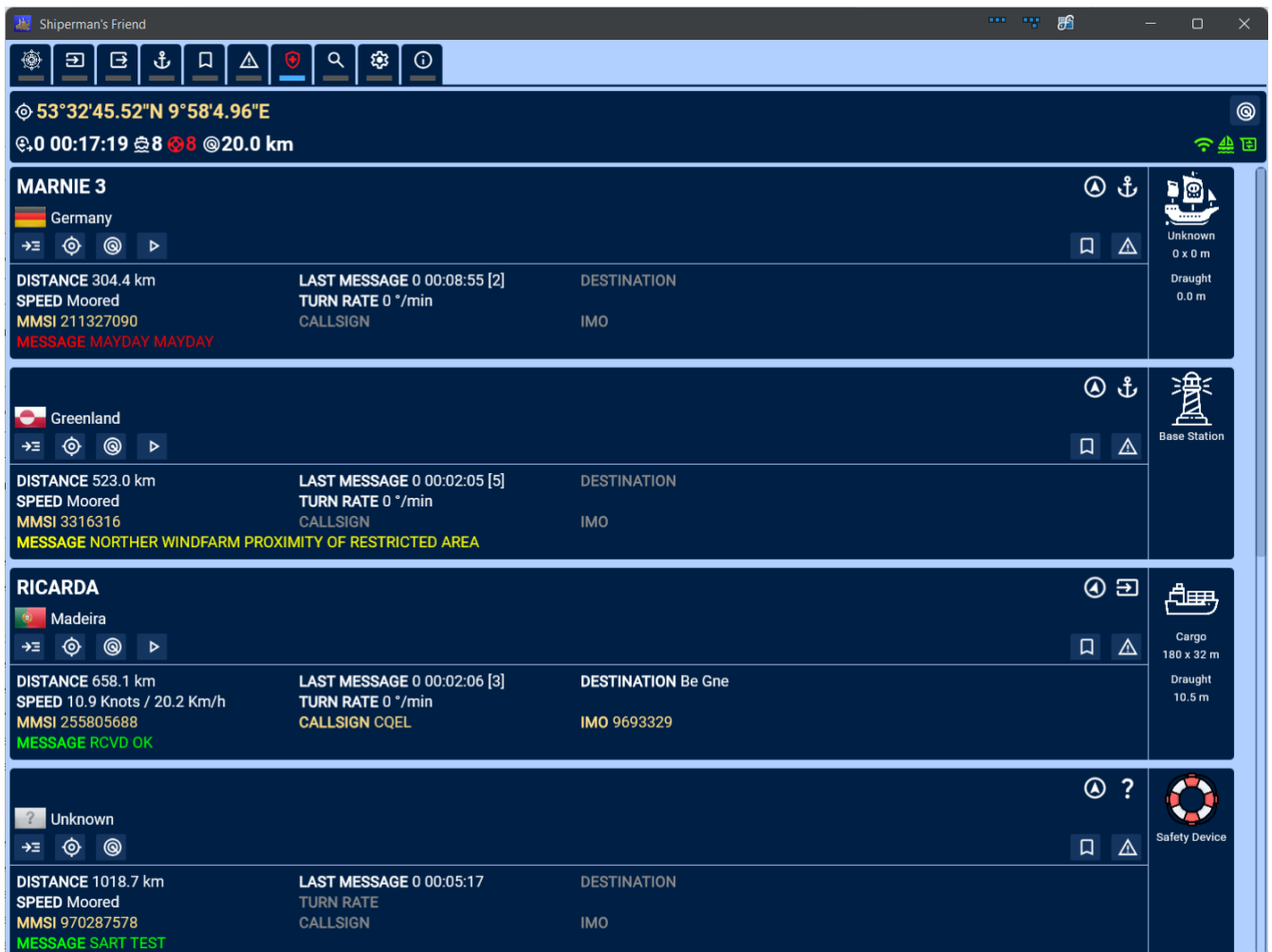
Perimeter Alert Tab

This tab shows all vessels you have selected for alertion when entering the inner bounding box (also depicted with y yellow ring on the radar page).



Safety Messages Tab

This tab shows currently active safety messages which have been collected within the outer bounds. Please note that these might not belong to your watched area. From the message text the severity of the message is determined using certain keywords for warnings and alerts. According to the severity the messages are also highlighted in red.



Search Vessels Tab

This tab allows you to search for vessel names, mmsi, imo, callsign, or ship category (english category names only). The overall layout is the same as for the driving vessels tab.



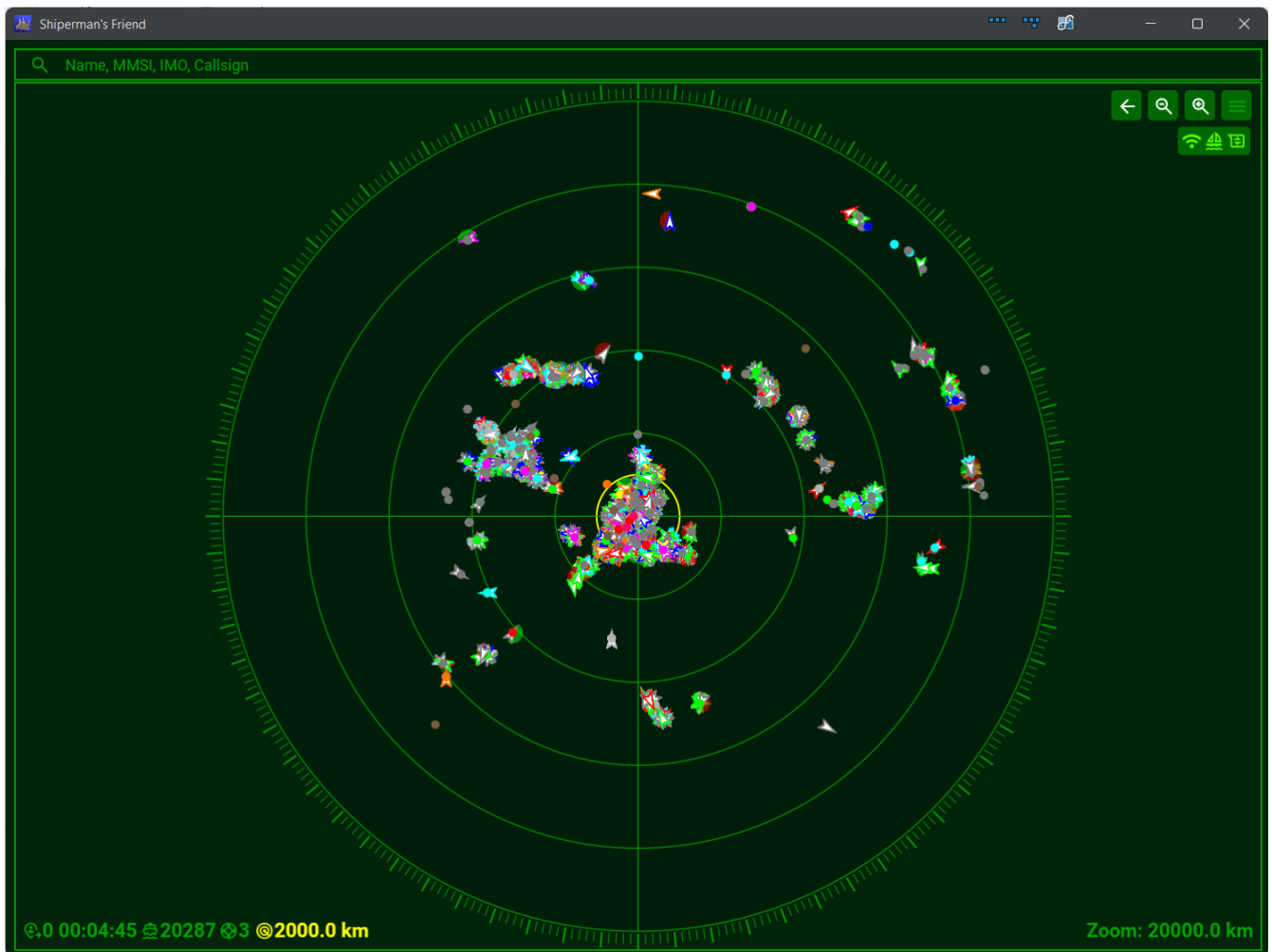
Radar Page

The radar page shows you all vessels within the inner radius to allow you to orientate in a dense spotting environment such as a harbor.

The currently selected vessel (that one where you clicked on the radar button) is highlighted with a white flashing circle. vessels having current safety messages are highlighted with a red flashing circle. Driving vessels are depicted with an arrow showing the current heading and moored vessels are displayed as a small circle by default (when the vessel is big enough its size as reported by AIS data is shown using a rectangle). The inner bounding box is depicted by a yellow circle when the zoom range is greater.

You can hover over vessel icons to show all vessel data in short form under the cursor or thumb.

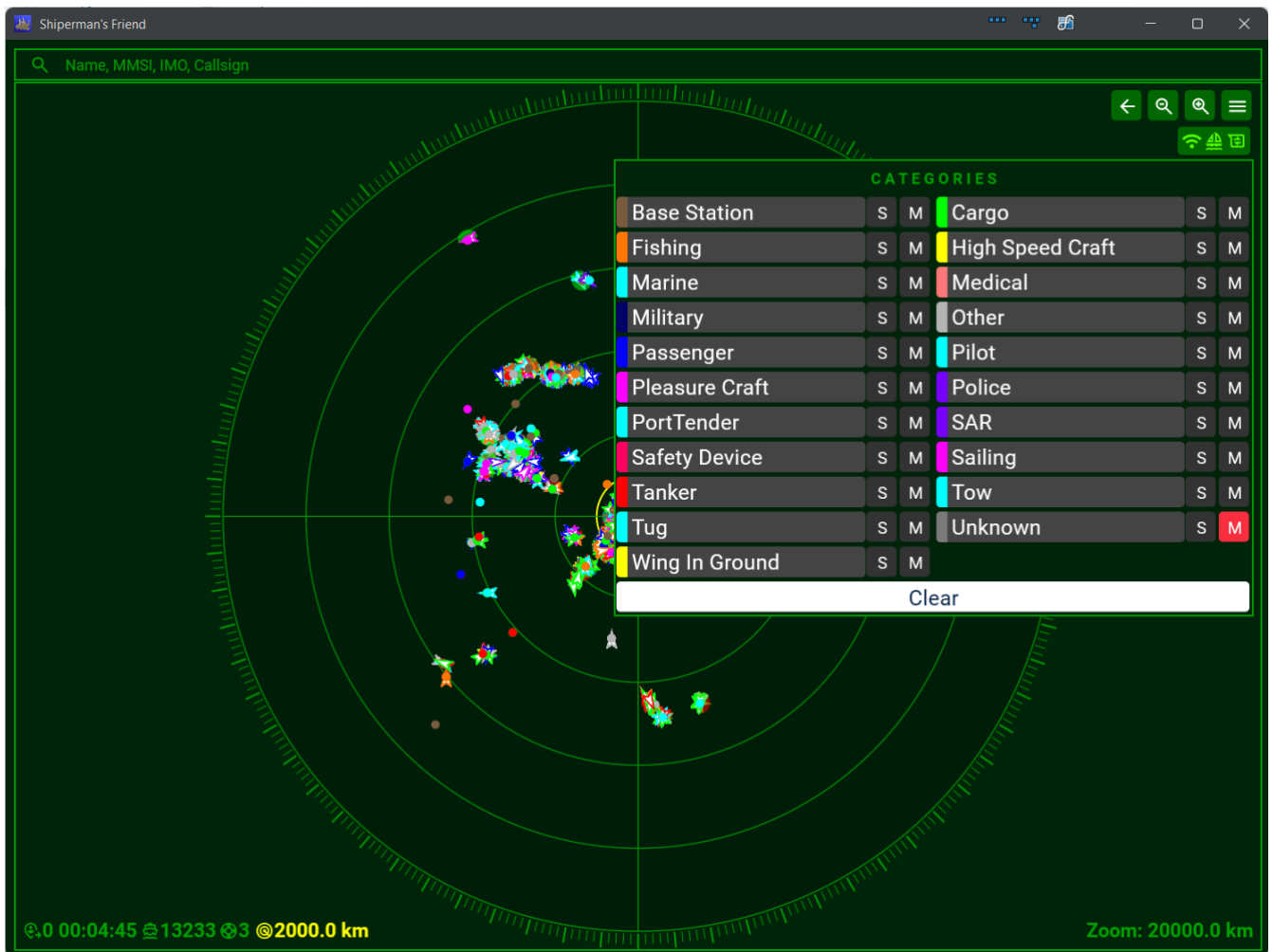
- In the top left corner you see the currently selected vessels name and below it's speed
- In the top right corner you find buttons to jump back to the previous view, zoom in and out and to open the category selector.
- In the bottom left corner the age of the current location, the number of visible vessels and the number of active safety messages are shown (in red when unread messages exist).
- In the bottom right corner you see the current zoom range



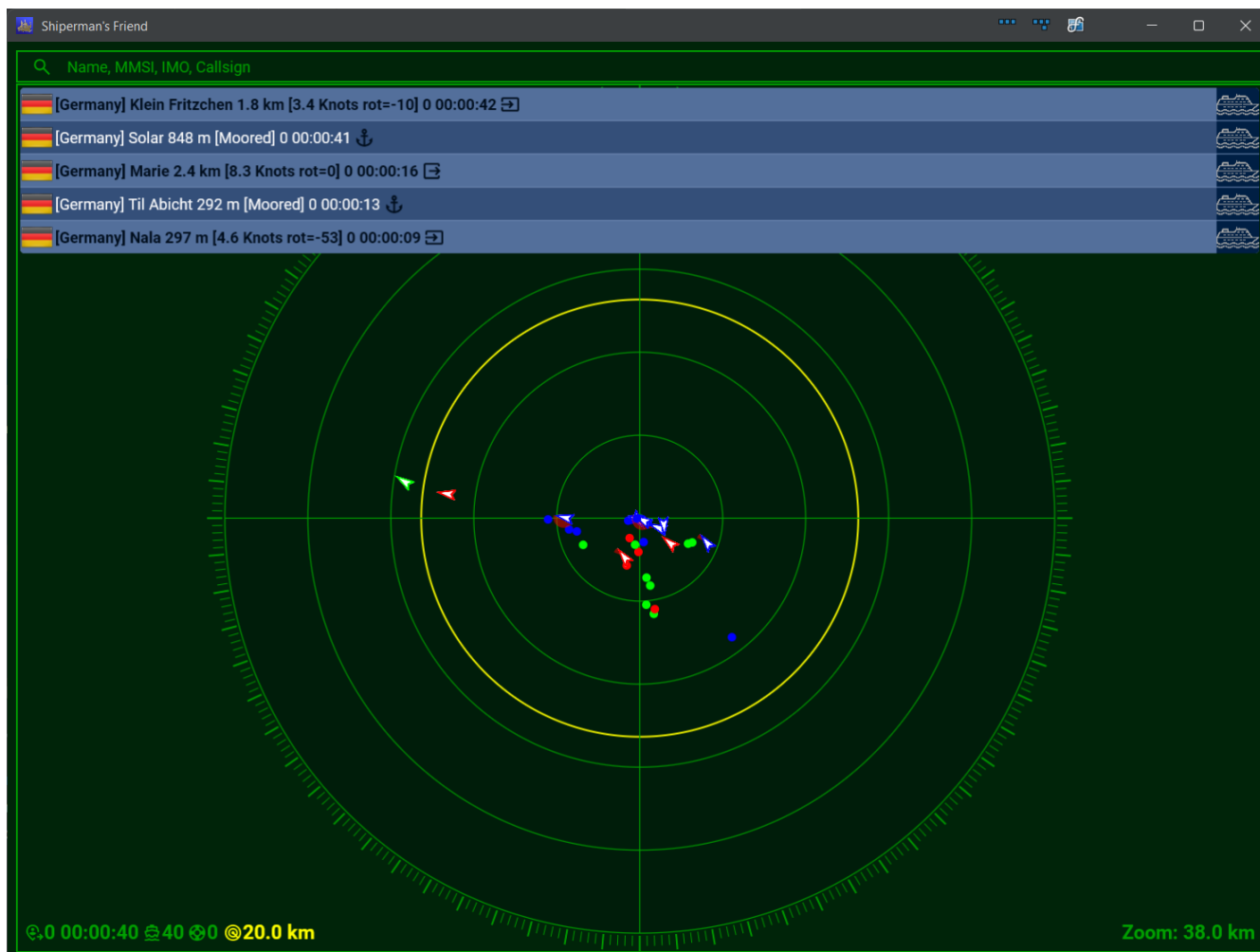
The shown vessels can be filtered by clicking on the burger menu on the top right which shows a flyover menu with all the known categories listed with their name, the standard category color and solo and mute buttons. Using the respective solo (S) or mute (M) buttons. You can either solo or mute categories. When you change the mode from solo to mute or vice versa all categories of the respective other mode will be cleared to avoid confusion what is muted or soloed. Use the clear button below to clear all the category selections.

These filters are combined with whatever you have entered in the top search bar.

The ship count in the top bar reflects the number of filtered vessels.



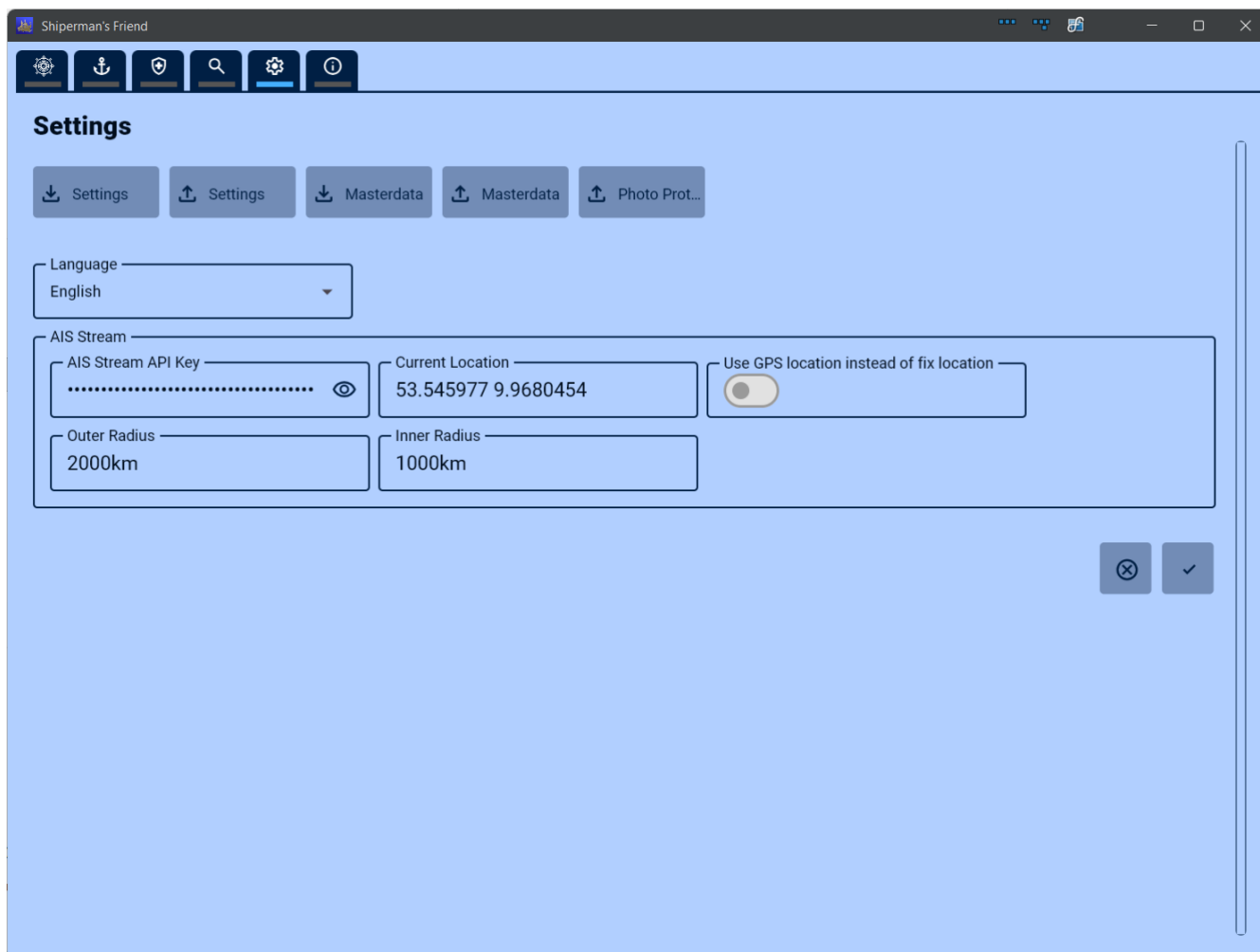
When you hover over a vessel with the mouse (or your finger on android) the most important infos for all vessels at that location is shown.



When there is a vessel which you have set up to be alerted for when it enters the inner bounds this vessel will be highlighted by a red pulsing ring.

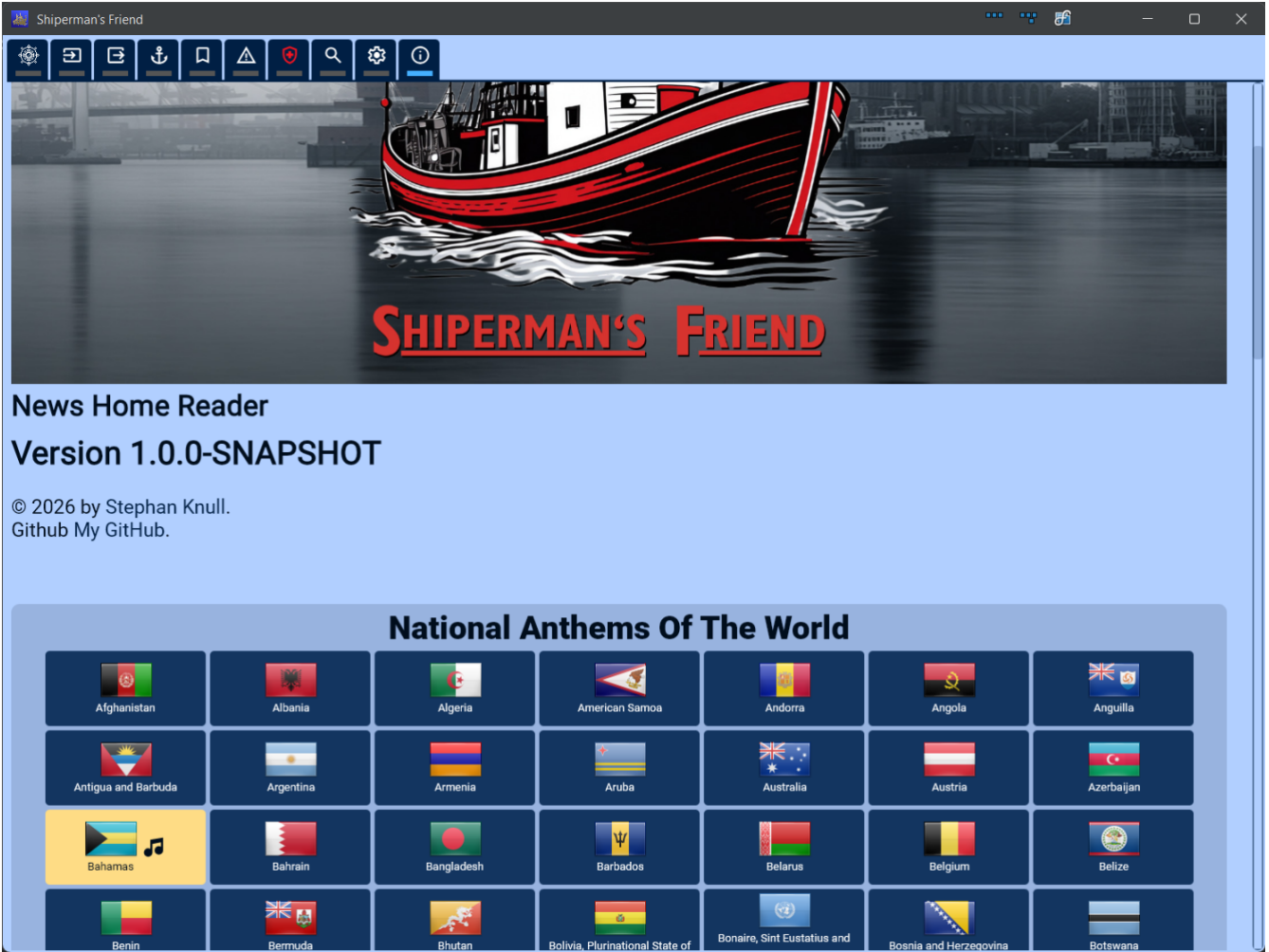
Settings Tab

The basic settings tab where you can set the outer and inner radius around the current location (when use gps location is switched on), or the entered coordinates as a fallback. The coordinates can be entered either as double values or in DMS notation. The radius can be entered either as double (not dot or fraction needed) or with either m (meters) or km (kilometers) unit. You can switch on and off "use gps location" during your session. The location will then be updated and used as the new centerpoint for your outer and inner bounding boxes. To obtain an API key for aisstream.io (which is currently the one and only free of charge open data source for AIS data) you have to register on their [website](#) and create an API key.



Info Tab

On this tab you can see the current version and contact infos. In the bottom part all countries are listed which have a national anthem available. when you click on it the respective anthem is played for you. So we have an anthem player included - ahoi!



Legal

License

[GNU GENERAL PUBLIC LICENSE v3 \(2007\)](#)

This project is licensed under GPLv3. If you use this code in your app, your entire app must also be open-sourced under GPLv3.

What this means for you

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Copyleft

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Attribution

Vessel Icons are taken from [Freepik - Flaticon](#)

Note

The banner was created by myself using Adobes Firefly AI engine which guarantees commercial compatibility. I did not use any original artwork of the well known suits company and researched that the Font used for the brand is indeed just Gill Sans which is available for free download in the internet. So no harm done here.